

My capstone project for my Data Technology course at Mountain Land Technical College will compare video game publishers. My data is called [Video Game Sales & Metacritic Intelligence 1980–2026](#) from Kaggle, and was created by Meruva Kodanda Suraj. Some important notes about this data are that all the country sales and global sales are in Millions. Another thing to keep note of is that any of the binary columns 1 is yes and 0 is no.

Next, I will go over some of the terminology. Some of the important ones that I use in my data are 100%, which means completing everything that there is to do in the game, and on the same note, completing the main campaign, which means playing the main part of the game without doing anything extra, for example, in a story game, there is the main story game play then there might be some side storys that you could play threw so saying you beat the main campaign would mean that you have not done any of the side storys. The next two I will be going over are the difference between platform type and just platform, so platform type is a generalization of what the game is played on, for example, if you played a game on a PC, on your phone, or on a Nintendo Switch, when referring to the type, this would mean with the example given, PC, a mobile,, or console games whereas when it just says platfor that is getting more specific like like naming out what you were playing the game on so with the same examples given earlier it would be still PC but with the phone game the would specify if the user was using a IOS phone or an Android phone to play the game same with the nintendo switch it would say nintendo switch or if they were playing games on like a nintendo wii it would say that nor just console.

One thing I would like to go over with this data is how much the publishers made across all the data and over the years, and compare platform-type sales to see which platform made the publisher the most money. This is important for getting a feel for the big names in the industry. This could also indicate whether there is a platform type that outsells the others, and which platforms have the lowest sales.

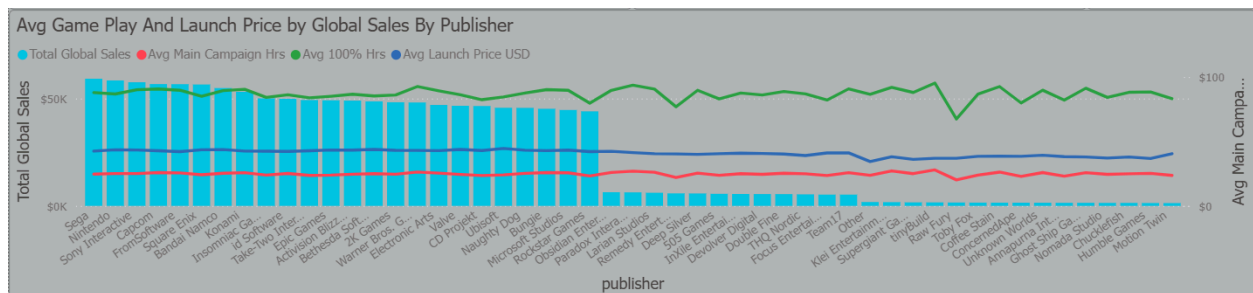
Another thing I would like to look into is how much gameplay a game has and how that affects its sales, or how much the game is when it is released. I would also like to look into some factors as well, like if the game is a sequel in a franchise or if the game has DLC (downloadable content, additional gameplay sold later after the main game is released), or if it is a game with online multiplayer (meaning multiple people from around the world can play the game together at the same time) if any of these factors could affect how long the game is or how much the game costs on release. This is good to look at; it could give some insight into what makes a sell or a flop, and the factors that could make it either way.

The Final thing I will be going over is how the user reviews compare to the critics' reviews of games by publisher, as well as what the ESRB (Entertainment Software Rating Board This is how people know if a game is for kids or adults, and in between) this could be very important because it show how most people feel about a game even if they were not meant to be the main audice for the game and how the critics scores differ form the normal persons reviews this also could show a tend of what types of games people are playing and what would be a good demographic to focus on when making a game.

Now I will go over how I got the data ready for use. To do this, I used Google BigQuery to create a project to store all my data and then perform the actual Data Cleaning and EDA (Exploratory data analysis). I used Python in Google Colab, connected to my Google BigQuery, and uploaded the results to my GitHub ([linked here](#)). The first thing I needed to do was make sure none of the data was missing. There was none, so there was no need to remove or fill in any empty rows. Next, I needed to make sure there were no duplicates, and there were none. Still, I did notice one thing, and that was that quite a few rows had overly long names, making the data harder to work with, so to save time, I renamed those rows by making copies of the original rows, but renamed them, so then I needed to make a new data frame without those duplicate old rows to keep things clean.

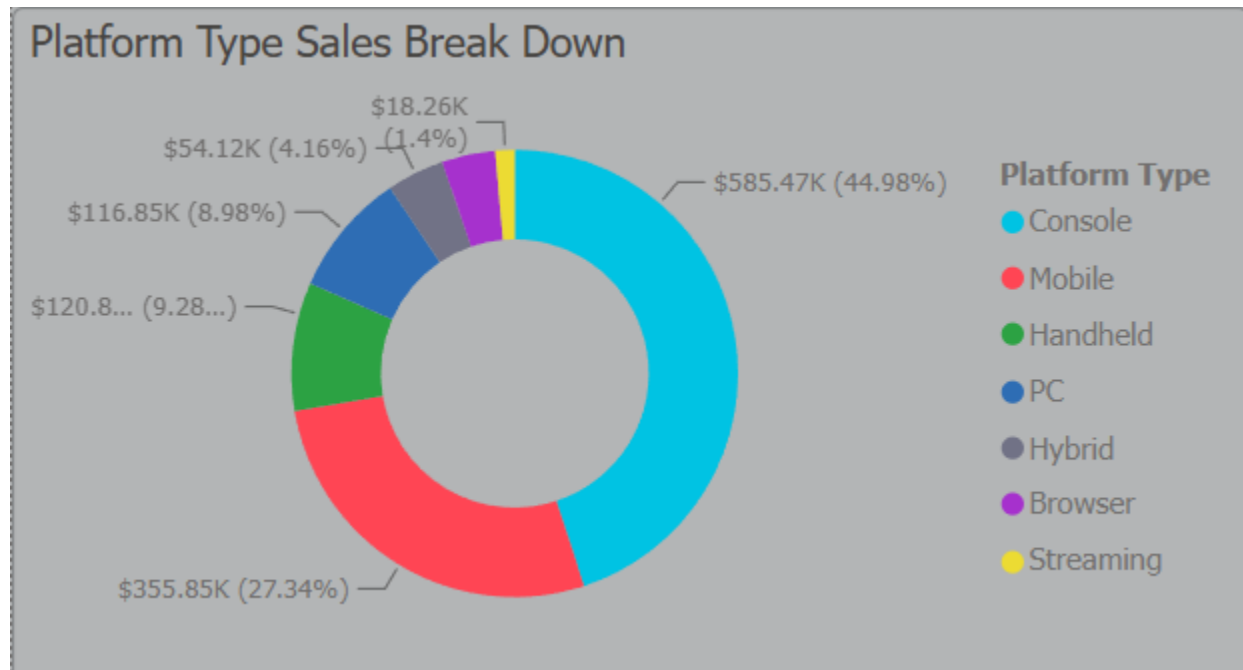
When I first started with this data, I assumed that most games would have been made for consoles, since they have been making games for consoles the longest, but I was wrong: PCs had the most games made for them. Another thing that I looked into was how many games were released each year. This is a good way to see if there was a year that was more popular for releasing games in, and the most popular years to release games are 2011 and 2019, but surprisingly, the third most popular was 2026, and this data only goes a few months into 2026, meaning that this year will be the year with the most games released.

Now that EDA was to get a feel for the spread of the data, I wanted to look a little more into the question that I wanted to answer, so one of the things I looked into was how the global sales compared to how long it takes on average to 100% a publisher's game to complete the main campaign and how much the on average the games cost on launch.



This graph shows that, for the most part, the average game price is fairly similar across the board. The average main campaign hours are less than the game's launch cost, but the time it takes to 100% the game is typically at least double the price. This does not affect how much money the publisher makes.

Another EDA graph I found interesting is the breakdown of the global sales by platform type.

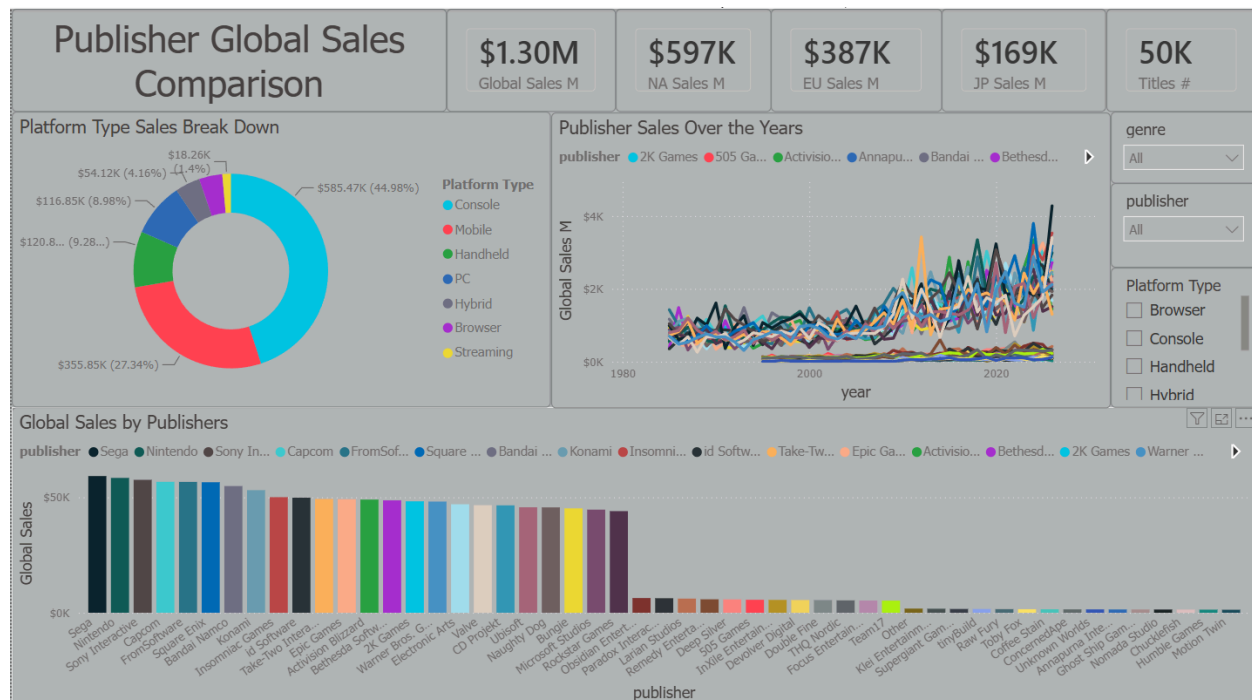


In this pie chart, you can see that consoles have accounted for the most over the years, with mobile games catching up but still lagging behind. In the near future, mobile games could surpass console games.

The last EDA chat I found interesting is this little chat that shows how many titles are in each ESRB rating, as well as how many times games in that rating were nominated and won Game of the Year. It also shows the average player review scores and Metacritic's average scores.

ESRB	# Titles	GOTY Nominated	GOTY Won	Avg User Score	Metacritic Score
T	1319	631	89	7.39	73.98
M	1308	632	103	7.42	74.11
E	1301	562	85	7.42	74.11
E10+	1189	244	42	7.40	73.93
Not Rated	977	140	16	7.40	74.06
AO	366	23	3	7.44	74.05

After my EDA, I created dashboards in Microsoft Power BI, and for my first dashboard, I wanted to review global sales for publishers.

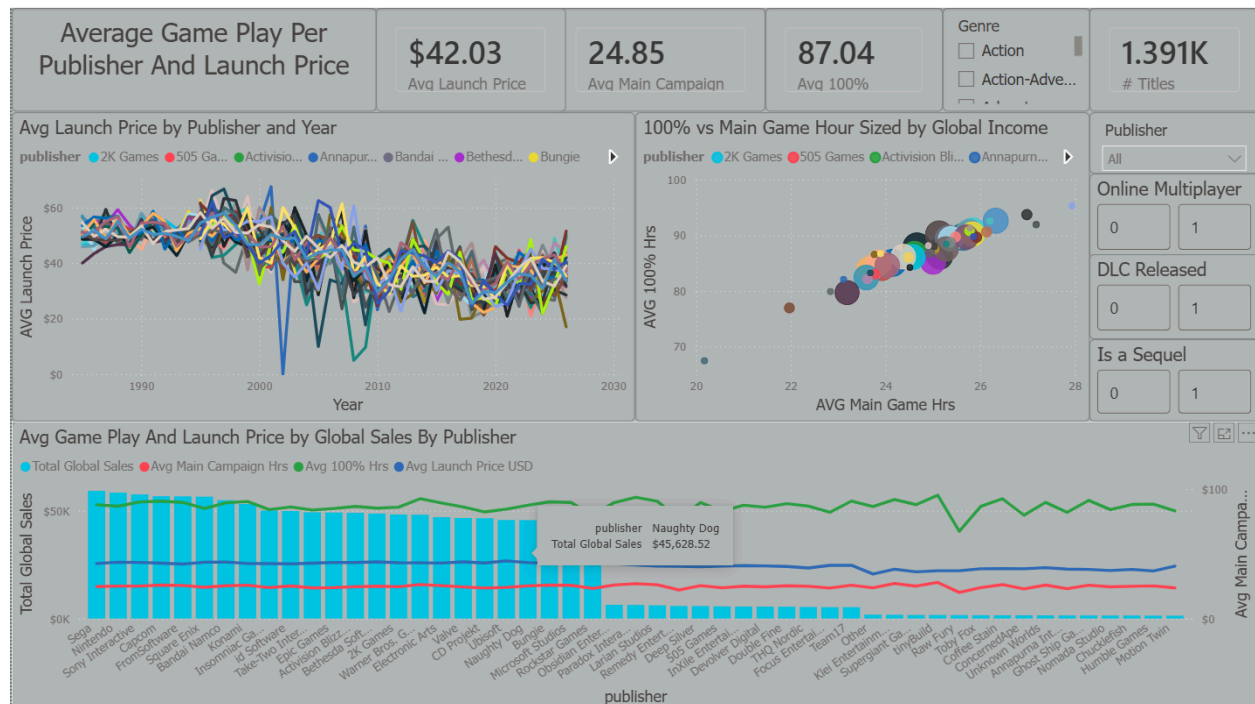


The biggest chart shows how much total the publisher has made in this data. This is a good way to see who the top publishers are, as well as the lowest-earning. This shows that SEGA and Nintendo are the top earners in this data, while Motion Twin and Humble Games are the lowest. You will notice two groups: the high earners and the low earners. All the publishers in each group are fairly close in earnings.

Now, when you pair this with the pie chart above, you can see which platform type had the highest earnings. Now, when you pair this with selecting specific publishers, you can see which platform made the most, and when you click through the publishers, you will see that most made their money from console games. This could be something we look into further in the future to understand why this is happening.

The final main graph on this dashboard shows how much the publisher made over the years, highlighting which years were good for them and which flopped. This is here to help you get a good understanding of how the publishers have performed over the years.

My next dashboard examines whether sales are affected by gameplay volume or launch price.

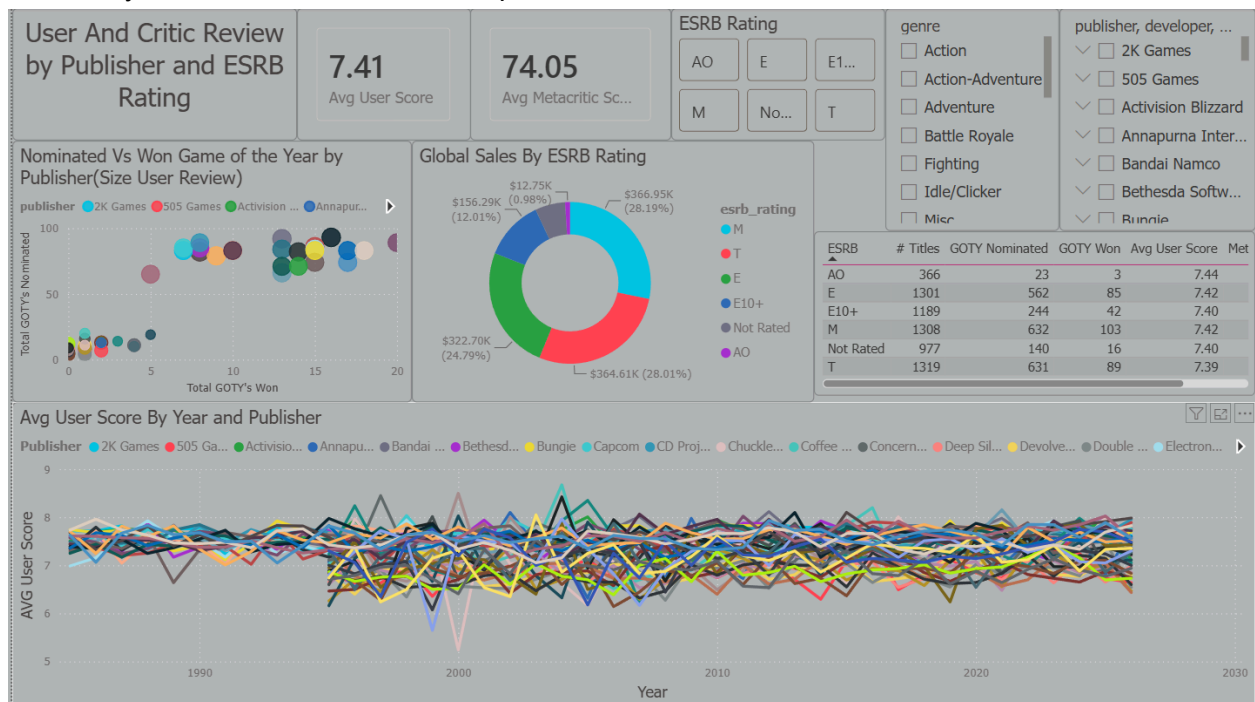


The main graph for this dashboard shows, as a line chart, the average number of hours it takes to beat the main campaign and 100% the game, and the average launch price, over how much each publisher made. Something you will notice, as mentioned earlier, is that for the most part, the averages for the main campaign and price are similar across the board, but what starts to differ from publisher to publisher is how long on average it takes to 100% one of their games some are quite a few hours more then others but all and all this dose not seem to affect how much each publisher made.

The next graph you might notice is the line chart showing the average launch price by year, with a line for each publisher, showing how much each publisher's average launch price changed each year. This shows how prices have changed over the years.

The final graph you will see is a dot plot comparing the time it takes to beat the main campaign vs. the time it takes to 100% the game, colored by publisher, with dot size indicating how much they made. This lets you see what appears to be a sweet spot for gameplay and profits.

My final dashboard covers the publishers' review and the ESRB.



The first graph in this dashboard is a line chart showing the average user score by year, with one line per publisher. This shows how people have rated the publisher over the years. This is a good way to show whether people were unhappy with the games they made that year or really liked the games they published that year.

The next graph you notice compares the nominated Game of the Year to the winner, and the size of the dots is proportional to the number of user reviews. This graph shows that most publishers that have won many GOTY awards have many reviews.

The next graph shows how much each game rating earned globally. This is a very good way to tell what type of game sold the most when looking at it. You see that M for mature games made the most, then T for teen, and after that E for everyone.

The Final chart you will see shows how many titles are in each ESRB rating, how many were nominated and won Game of the Year, and, finally, the average user and Metacritic ratings when paired with all the other graphs. This is a good way to see what type of games won the most awards.

So, in conclusion, when looking at the dashboards, you may be able to tell what a good type of game to produce would be, for example, with all the things that I have pointed out, a game that is on a console and mobile that is rated M or T and published by SEGA could be fairly profitable. Still, it would be best to look into additional factors, like genre, sequels, and DLC, and delve deeper into them to help you reach a more profitable game.